On the Closed-Form of Flow Matching

Generalization Does Not Arise from Target Stochasticity

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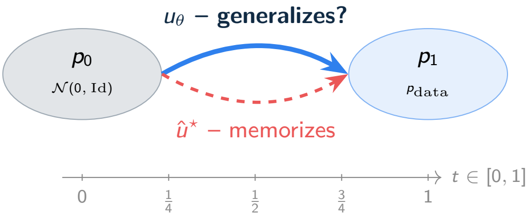
Paper: Bertrand, Gagneux, Massias, Emonet. *NeurIPS 2025.*

- Diffusion models and flow matching are state-of-the-art
- Both define a path $(p_t)_{t\in[0,1]}$: noise $\xrightarrow{u_\theta}$ data
- Optimal empirical velocity $\hat{u}^\star$ has a **closed-form** expression from finite data

Central paradox

Solving the ODE with $\hat{u}^\star$ can **only** produce training data points.

$\Rightarrow$ *How can FM generalize if its optimal velocity memorizes?*



Prevailing hypothesis (Vastola, ICLR 2025): generalization $\leftarrow$ stochasticity of u^{cond}.

This paper rigorously disproves it.

Forward process (SDE)

$$\mathrm{d}X_t = f(X_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t, \quad X_0 \sim p_{\mathrm{data}}$$

Linear SDE: $X_t | X_0 \sim \mathcal{N}(\alpha_t X_0, \sigma_t^2 \mathrm{Id})$

Reverse-time SDE

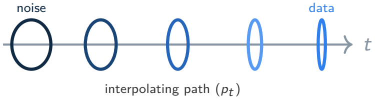
$$\mathrm{d}\bar{X}_\tau = \Big[-f + g^2 \underbrace{\nabla \log p_{T-\tau}(\bar{X}_\tau)}_{\text{score}} \Big] \mathrm{d}\tau + g\,\mathrm{d}\bar{W}_\tau$$

Score matching loss: $\mathcal{L}_{\mathrm{DSM}} = \mathbb{E}\left[\left\| s_\theta(\alpha_t x_0 + \sigma_t \varepsilon, t) + \frac{\varepsilon}{\sigma_t} \right\|^2 \right]$

Probability-flow ODE (bridge to FM)

$$\frac{\mathrm{d}}{\mathrm{d}t}\tilde{X}_t = f(\tilde{X}_t, t) - \frac{g(t)^2}{2} \nabla \log p_t(\tilde{X}_t)$$

- Same marginals p_t as the SDE
- Fully **deterministic** – no noise at inference
- Direct bridge to Flow Matching



The FM framework

Learn $u : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ such that

$$\dot{x}(t) = u(x(t), t), \quad x(0) \sim \mathcal{N}(0, \mathrm{Id})$$

transports p_0 to p_{data}. Equivalently: (p_t, u) satisfies the continuity equation $\partial_t p_t + \nabla \cdot (p_t u) = 0$.

Conditional (Gaussian) path

$$p_t(\cdot|x_1) = \mathcal{N}\big(tx_1, (1-t)^2 \mathrm{Id}\big)$$

with conditional velocity $u^{\mathrm{cond}}(x, x_1, t) = \frac{x_1 - x}{1-t}$.

CFM loss

$$\mathcal{L}_{\mathrm{CFM}}(\theta) = \mathbb{E}\Big[\|u_\theta(x_t, t) - (x_1 - x_0)\|^2\Big]$$

where $x_t = (1-t)x_0 + tx_1$.

Variance decomposition lemma

For any A, X: $\mathbb{E}[\|f(X) - A\|^2] = \mathbb{E}[\|f(X) - \mathbb{E}[A|X]\|^2] + \underbrace{\mathbb{E}[\|A - \mathbb{E}[A|X]\|^2]}_{\text{indep. of } \theta}$

$\Rightarrow$ $\mathcal{L}_{\mathrm{CFM}}$ and the FM loss share the **same gradient and minimizer**:

$$u^\star(x, t) = \mathbb{E}_{z|x,t}\big[u^{\mathrm{cond}}(x, z, t)\big].$$

Theoretical Equivalence: FM $\equiv$ Diffusion | Both learn the same probability-flow ODE drift

Theorem 1 (Lipman et al. 2024, Thm. 5.1)

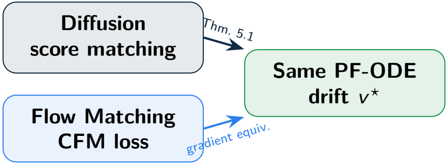
Under affine Gaussian paths $x_t = \sigma_t \varepsilon + \alpha_t x_0$, the FM optimal velocity equals the PF-ODE drift:

$$\boxed{v^\star(x,t) = f_t\, x - \frac{g_t^2}{2}\, \nabla \log p_t(x)}$$

where $f_t = \dot\alpha_t/\alpha_t$, $g_t^2 = 2\sigma_t\dot\sigma_t - 2f_t\sigma_t^2$.

Proof sketch. Write $v^\star(x,t) = \mathbb{E}[\dot x_t | x_t = x]$. Gaussian posterior identities give $\mathbb{E}[\varepsilon|x] = -\sigma_t s$, $\mathbb{E}[x_0|x] = \frac{1}{\alpha_t}(x + \sigma_t^2 s)$ (with $s = \nabla \log p_t$). Substituting and identifying f_t, g_t yields the result. $\square$

Canonical schedule $\alpha_t = t, \sigma_t = 1 - t$: $u^{\text{cond}} = x_1 - x_0$, the simplest form.



Key takeaway

FM and diffusion are **mathematically equivalent**. Generalization *in practice* is not explained by equivalence alone.

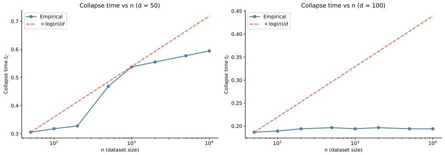
Replace p_{data} by $\hat{p} = \frac{1}{n}\sum_{i=1}^{n} \delta_{x^{(i)}}$.

Proposition 1 (Bertrand et al. 2025)

$$\hat{u}^{\star}(x,t) = \sum_{i=1}^{n} \lambda_i(x,t)\, \frac{x^{(i)} - x}{1-t}$$

$$\lambda_i = \mathrm{softmax}\Big(-\frac{\|x - t x^{(j)}\|^2}{2(1-t)^2}\Big)_j$$

As $t \to 1$: one weight $\lambda_i \to 1 \Rightarrow \hat{u}^{\star}$ points to a **single training sample**.



Collapse time t_C vs. n at $d \in \{50, 100\}$. Follows $\propto \log(n)/d$ (dashed).

The central paradox

$\hat{u}^{\star}$ memorizes — yet u_θ generalizes.

Hypothesis (Vastola): stochasticity of u^{cond}.

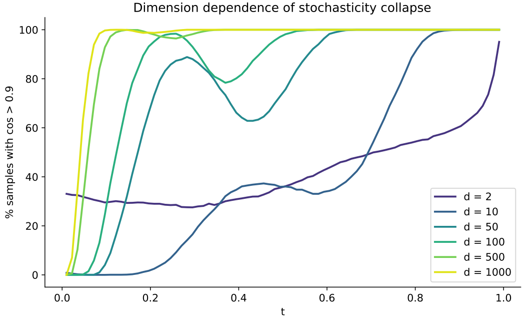
This paper: limited capacity. ✓

Collapse time (Biroli et al. 2024)

$$t_C \sim \mathcal{O}\left(\frac{\log n}{d}\right)$$

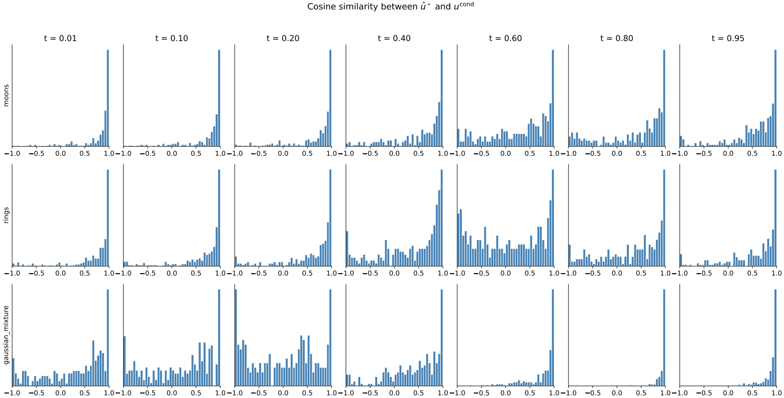
In high d: $t_C \ll 1$, stochastic phase **vanishingly short**.

Setting	d	n	t_C
2D toy	2	2000	3.8
d=100	100	2000	0.075
CIFAR-10	3072	50k	0.004

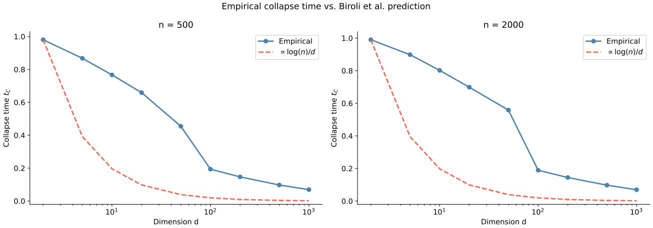


Fraction of pairs with $\cos(\hat{u}^\star, u^{\mathrm{cond}}) > 0.9$ vs. t, $d \in \{2, 10, 50, 100, 500, 1000\}$, $n = 2000$. Higher dimension $\Rightarrow$ earlier collapse.

Figure 1a — Cosine Similarity Histograms | $\cos(\hat{u}^\star, u^{\text{cond}})$ at 7 time steps for 3 datasets ($n = 5000$)



Each row: one 2D dataset (moons, rings, Gaussian mix.), each column: one time step. Histograms **concentrate near 1** as $t \to 1$ — softmax collapses to a single training point. In high d, this collapse occurs at $t < 0.1$.



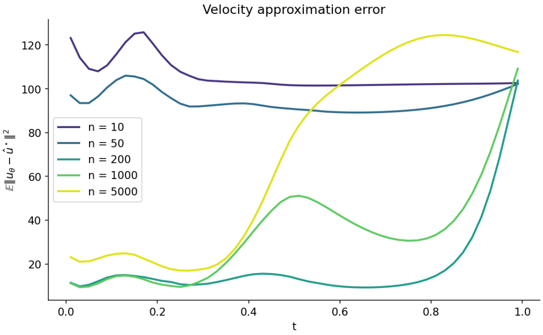
Empirical t_C (blue) vs. theoretical $\propto \log(n)/d$ (dashed red) for $n = 500$ and $n = 2000$.

Key findings:

- Fit is excellent across $d \in [2, 10^3]$
- Both $n = 500$ and $n = 2000$ follow the same law
- The Biroli et al. prediction, originally derived for *diffusion* models, **transfers to flow matching**

The stochastic phase covers $<$ 4% of the trajectory at $d = 3072$ (CIFAR-10 resolution).

Figure 2 — Velocity Approximation Failure | The network fails to approximate $\hat{u}^\star$ at two critical time intervals



$\mathbb{E}\,\|u_\theta - \hat{u}^\star\|^2$ vs. t for the $d = 100$ Gaussian mixture, varying n.

Two failure zones:

1. **Early bump** ($t \approx 0.12$–0.15): stochastic-to-deterministic transition; coincides with t_C at $d = 100$.
2. **Late rise** ($t \to 1$): $\hat{u}^\star$ becomes non-Lipschitz; error grows with n.

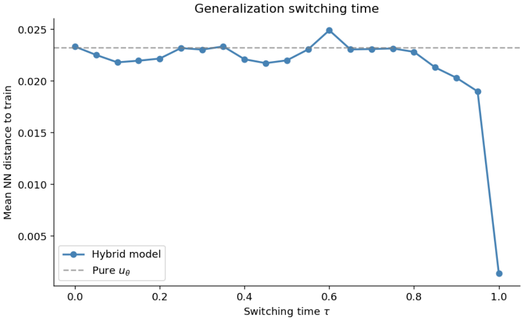
Causal chain:

$t < t_C$: $u_\theta \approx u^{\mathrm{cond}} \neq \hat{u}^\star \Rightarrow$ **generalization**.

$t > t_C$: $u^{\mathrm{cond}} \approx \hat{u}^\star \Rightarrow$ small error.

NN distance $\uparrow$ as $n \uparrow$: large dataset forces the model to generalize (capacity bottleneck).

Figure 3 — Generalization Switching Time | The first ∼20% of the trajectory determines generalization



Mean NN distance vs. switching time τ for the 2D Gaussian mixture ($n = 2000$). Hybrid model: follow $\hat{u}^\star$ on $[0, \tau]$, then u_θ on $[\tau, 1]$.

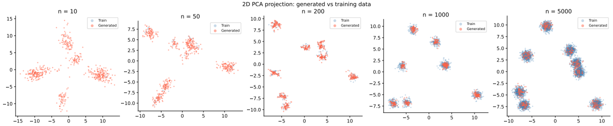
Setup: hybrid sampler on 2D Gaussian mix., $n = 2000$.

Results:

- **Plateau** $\tau \leq 0.60$: same NN distance as pure u_θ — generalization intact
- **Sharp drop** at $\tau \approx 0.85$–0.95: replacing the full trajectory by $\hat{u}^\star$ collapses generalization
- At $\tau = 1.0$: pure memorization

Main finding

Generalization lives where $u_\theta \not\approx \hat{u}^\star$. Using $\hat{u}^\star$ in that window eliminates it entirely.



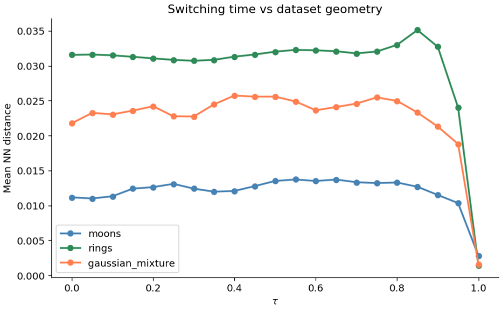
2D PCA projections of generated (red) vs. training (blue) samples for $n \in \{10, 50, 200, 1000, 5000\}$, $d = 100$.

- **Small** n ($n = 10$): generated points cluster *on* training data $\Rightarrow$ memorization
- **Large** n ($n = 5000$): generated points form a broader distribution preserving modal structure $\Rightarrow$ generalization

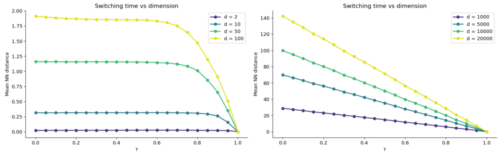
Interpretation

As n grows beyond the network capacity, $\hat{u}^\star$ becomes too complex to approximate $\Rightarrow$ the model **must** generalize.

Beyond the Paper — Geometry & Dimension Scaling | How manifold structure and ambient dimension modulate generalization

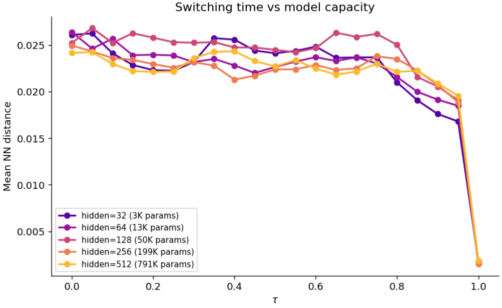


Switching curves for 3 geometries ($n = 2000$). Moons (structured, low intrinsic dim) $\rightarrow$ lowest NN distance. Complexity of $\hat{u}^\star$ depends on the data manifold.

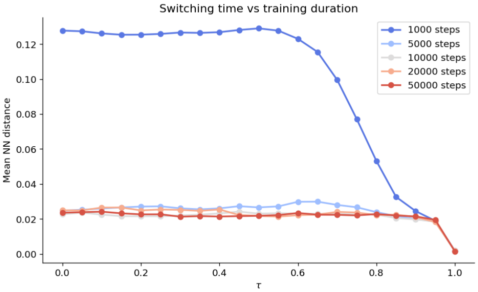


Left: $d \in \{2, 10, 50, 100\}$ — switching shifts earlier as d grows. Right: $d \in \{1000, \ldots, 20000\}$ — curve becomes *flat*: $\hat{u}^\star$ itself generalizes (mem./gen. dichotomy **vanishes**).

At very high d: $t_C \approx \log(n)/d \rightarrow 0$, so $\hat{u}^\star \approx u^{\mathrm{cond}}$ everywhere and the closed-form field already generalizes by necessity.



Hidden dim $\in \{32, 64, 128, 256, 512\}$ ($d = 2$, Gaussian mix., $n = 2000$). Larger networks approximate $\hat{u}^\star$ better $\Rightarrow$ switching time shifts to smaller τ. hidden$= 32$: never fully memorizes.

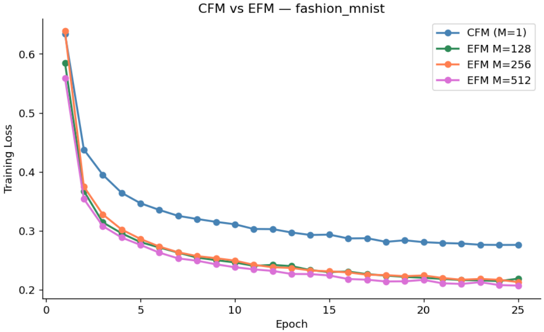


$\{1k, \ldots, 50k\}$ gradient steps. Under-trained (1k): flat, strong generalization. At 50k: converges; bottleneck is **capacity**.

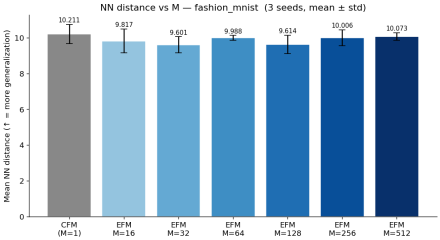
Consistent message

More capacity & training $\rightarrow$ better $\hat{u}^\star$ approximation $\rightarrow$ **more memorization**.
Opposite of the prevailing intuition.

EFM on Fashion-MNIST | Reducing target stochasticity does not improve generalization



Training loss (CFM vs. EFM for $M \in \{128, 256, 512\}$). EFM achieves significantly lower loss.



NN distance vs. M (3 seeds): flat.

- EFM replaces u^{cond} by $\hat{u}^{\star}_{M}$ (M samples per step)
- Lower training loss, **identical generalization**

Target variance does **not** drive generalization.
Root cause: **limited network capacity**.

Robust findings

1. $t_C \sim \log(n)/d$ transfers from diffusion to the FM setting
2. Generalization $\leftrightarrow$ large $\mathbb{E}\,\|u_\theta - \hat{u}^\star\|^2$ at early/late t
3. More capacity / more training $\Rightarrow$ memorization
4. EFM confirms: noise $\not\Rightarrow$ generalization

Critical perspective

- **(i)** Collapse result theoretically expected (gradient equiv. + measure concentration)
- **(ii)** EFM tension: variance reduction only affects the stochastic phase ($< 15\%$ of trajectory at $d = 100$)
- **(iii)** True contribution: pinpointing *when* generalization occurs via the hybrid model

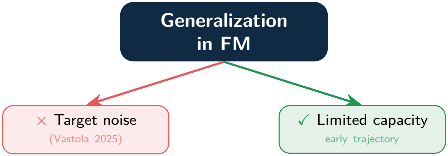
Open questions

- **Inductive biases:** U-Net vs. MLP radically changes $\hat{u}^\star$ approximation
- **Very high d:** does $\hat{u}^\star$ itself generalize at image resolution?
- **Data augmentation** and effective complexity of $\hat{u}^\star$
- **LLM memorization:** analogous dynamics in token flows?

Broader implication: Scaling capacity may be at odds with generalization. Yet large models generalize on natural images — suggesting **effective complexity** of $\hat{u}^\star$ (not raw capacity) is the key quantity.

Summary

1. **FM $\equiv$ Diffusion:** same PF-ODE drift; four prediction targets algebraically equivalent.
2. **$\hat{u}^\star$ non-stochastic in high dims:** stochastic phase $\sim \log(n)/d$ — verified on toy datasets across two orders of magnitude in d.
3. **Stochasticity $\not\Rightarrow$ generalization:** EFM lowers training loss but not NN distance.
4. **Generalization $\Leftarrow$ limited capacity:** failure to approximate $\hat{u}^\star$ at early times is the true mechanism.
5. **Reproduced & extended:** Figs. 1–3 confirmed; ablations on geometry, dimension, capacity, training all consistent.



All experiments on a single Google Colab T4 GPU in <30 min/notebook.

References

1. **Bertrand, Gagneux, Massias, Emonet.** *On the closed-form of flow matching.* NeurIPS 2025.
2. **Biroli, Bonnaire, de Bortoli, Mézard.** *Dynamical regimes of diffusion models.* Nature Comm. 15(1):9957, 2024.
3. **Lipman, Chen, Ben-Hamu, Nickel, Le.** *Flow matching for generative modeling.* ICLR 2023.
4. **Lipman et al.** *Flow matching guide and code.* arXiv:2412.06264, 2024.
5. **Liu, Gong, Liu.** *Flow straight and fast: rectified flow.* ICLR 2023.
6. **Albergo, Vanden-Eijnden.** *Building normalizing flows with stochastic interpolants.* ICLR 2023.
7. **Ho, Jain, Abbeel.** *Denoising diffusion probabilistic models.* NeurIPS 2020.
8. **Song et al.** *Score-based generative modeling through SDEs.* ICLR 2021.
9. **Vastola.** *Generalization through variance.* ICLR 2025.
10. **Gao, Li.** *How do FM models memorize and generalize?* arXiv:2410.23594, 2024.

Under affine Gaussian paths $x_t = \sigma_t \varepsilon + \alpha_t x_0$, write $\dot{x}_t = \sigma'_t \varepsilon + \alpha'_t x_0$.

Step 1. $v^\star(x,t) = \mathbb{E}[\dot{x}_t | x_t = x] = \sigma'_t \mathbb{E}[\varepsilon|x] + \alpha'_t \mathbb{E}[x_0|x]$.

Step 2. Gaussian posterior mean identities (setting $s = \nabla \log p_t$):

$$\mathbb{E}[\varepsilon|x] = -\sigma_t s(x,t), \qquad \mathbb{E}[x_0|x] = \tfrac{1}{\alpha_t}(x + \sigma_t^2 s(x,t)).$$

Step 3. Substituting:

$$v^\star(x,t) = \frac{\dot{\alpha}_t}{\alpha_t} x + \left(\frac{\dot{\alpha}_t}{\alpha_t} \sigma_t^2 - \dot{\sigma}_t \sigma_t \right) s(x,t).$$

Step 4. Setting $f_t = \dot{\alpha}_t / \alpha_t$ and $g_t^2 = 2\sigma_t \dot{\sigma}_t - 2 f_t \sigma_t^2$ gives $(\cdots) = -g_t^2/2$, so:

$$\boxed{v^\star(x,t) = f_t\, x - \tfrac{g_t^2}{2} \nabla \log p_t(x)} \qquad \square$$

EFM loss: replace u^{cond} by a Monte Carlo estimate $\hat{u}^\star_M$ with M samples:

$$\mathcal{L}_{\mathrm{EFM}}(\theta) = \mathbb{E}\left[\left\|u_\theta(x_t, t) - \hat{u}^\star_M(x_t, t)\right\|^2\right]$$

where $b^{(1)} := x_1$ (the conditioning point) and

$$\hat{u}^\star_M(x, t) = \sum_{j=1}^{M} \frac{b^{(j)} - x}{1 - t} \cdot \mathrm{softmax}\left(-\frac{\|x - tb^{(k)}\|^2}{2(1-t)^2}\right)_{k,j}.$$

Key property: including x_1 as $b^{(1)}$ makes $\hat{u}^\star_M$ *unbiased* (Prop. 2, Bertrand et al.) with lower variance. As $M \to \infty$: $\hat{u}^\star_M \to \hat{u}^\star$ pointwise.

Fashion-MNIST: EFM achieves strictly lower training loss but **identical NN distances** across all $M \in \{1, ..., 512\}$. This definitively rules out stochasticity as a driver of generalization.

Softmax dominance condition: weight $\lambda_{i^\star}$ dominates when the gap in the exponent exceeds $\log n$, i.e. $\frac{\|x_t - tx^{(i^\star)}\|^2 - \|x_t - tx^{(j)}\|^2}{2(1-t)^2} \gg \log n$ for all $j \neq i^\star$.

Scaling of squared distances: For i.i.d. Gaussian data in $\mathbb{R}^d$, pairwise distances concentrate around $\sim d$ (law of large numbers). Along $x_t = (1-t)x_0 + tx_1$, the nearest-neighbour gap scales as $\sim d \cdot t^2$. Solving gap $\sim d \cdot t_C = \log n$ gives:

$$\boxed{t_C \approx \frac{\log n}{d}}$$

CIFAR-10 example ($d = 3072$, $n = 50\,000$):

$$t_C \approx \frac{\log 50000}{3072} \approx 0.036 \quad \Rightarrow \quad \text{stochastic phase} < 4\%.$$

The Biroli et al. result, proved for diffusion models via a careful SDE analysis, holds empirically for FM — confirmed across two orders of magnitude in d on our toy datasets.

Appendix D — Ablation Summary

Variable	Effect	Mechanism	Interpretation
$\uparrow$ dim. d	$\tau^\star \downarrow$	$t_C \propto 1/d$	non-stochastic regime starts earlier
$\uparrow$ size n	stable	capacity bottleneck	same failure zone
$\uparrow$ capacity	$\tau^\star \downarrow$	better $\hat{u}^\star$ approx.	failure window shrinks
$\uparrow$ training	$\tau^\star \downarrow$	better $\hat{u}^\star$ approx.	bottleneck: capacity not opt.
Geometry	varies	$\hat{u}^\star$ complexity	moons $\rightarrow$ earliest switching
$\uparrow$ M (EFM)	no change	noise $\not\rightarrow$ gen.	central claim confirmed
$d \geq 1000$	flat curve	$\hat{u}^\star$ generalizes	dichotomy vanishes

All ablations consistently support: generalization $\Leftrightarrow$ poor $\hat{u}^\star$ approximation at early times, driven by limited network expressiveness.